

Development of an Intelligent Multi-Modal Weather Intelligence and Climate Decision Support Platform Integrating Machine Learning, Deep Learning, Remote Sensing, and Agentic Large Language Models

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Abstract—The Indian Southwest Monsoon System is the socio-economic backbone of South Asia, influencing agricultural yields, food security, hydroelectric capacity, and regional disaster preparedness. However, predicting its spatio-temporal dynamics and translating predictions into actionable, localized policy decisions remains a grand challenge. This research presents an end-to-end, open-source Multi-Modal Weather Intelligence and Climate Decision Support Platform. The system bridges quantitative forecasting with qualitative decision support across two core phases. Part 01 constructs a high-precision predictive engine by integrating statistical, machine learning (XGBoost, CatBoost), deep learning (Bi-LSTM, Temporal Fusion Transformer), and remote sensing vision models (U-Net, SAM, YOLOv8) trained on IMD, ERA5, and Sentinel-2 data. Part 02 transforms this predictive core into an agentic Generative AI system using open-weight Large Language Models (LLMs), LangGraph, Vector Databases (FAISS/ChromaDB), Retrieval-Augmented Generation (RAG), and the Model Context Protocol (MCP). The proposed architecture enables multi-step autonomous reasoning, interactive model invocation, and climate report generation without reliance on proprietary APIs or paid commercial software.

Index Terms—Generative AI, Indian Southwest Monsoon, Retrieval-Augmented Generation (RAG), AI Agents, Remote Sensing, Time-Series Forecasting, LangGraph, Model Context Protocol (MCP), Deep Learning.

I. INTRODUCTION

The Indian Southwest Monsoon (June to September) supplies over 70% of South Asia’s annual precipitation [1]. Variations in its onset, spatial distribution, and localized intensity routinely lead to extreme weather events such as flash floods, severe droughts, and landslides [1]. Traditional Numerical Weather Prediction (NWP) models, while physics-grounded, demand extensive supercomputing clusters and often fail to present accessible, context-aware insights for non-technical domain stakeholders, regional policy planners, and disaster management authorities [1].

To address these multi-faceted challenges, the MacroEdtech GenAI Research Initiative established a

project framework aimed at unifying statistical predictive modeling with agentic Large Language Model (LLM) workflows [1]. This paper documents the theoretical, empirical, and architectural development of the resulting platform [1].

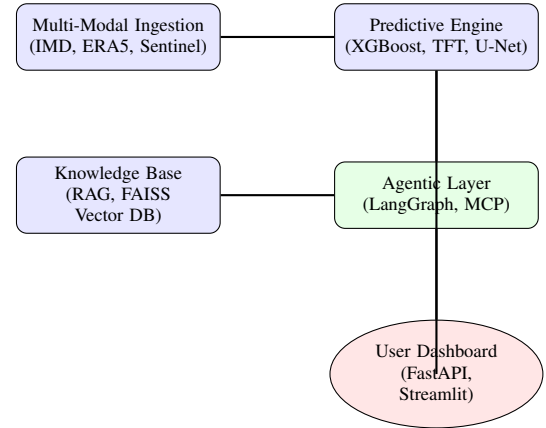


Fig. 1. End-to-End Conceptual Flow Diagram of the Multi-Modal Weather Platform.

The primary contributions of this paper are organized as follows:

- 1) **Data Pipelines & Remote Sensing Integration:** Acquisition, spatial transformation, and feature engineering across satellite imagery (Sentinel-2, Landsat, MODIS) and reanalysis datasets (ERA5 [2], IMD) [1].
- 2) **Comprehensive Predictive Benchmarking:** Evaluation of regression algorithms (XGBoost [3]), deep sequence architectures (Bi-LSTM, GRU), and spatio-temporal models (Temporal Fusion Transformer [4]) [1].
- 3) **Vision-Based Satellite Analysis:** Computer vision segmentation and detection workflows for cloud profiling

and flood risk mapping using U-Net [5], SAM [6], and YOLOv8 [1].

- 4) **Agentic GenAI & RAG Architecture:** A multi-agent orchestration layer built on LangGraph, vector databases, RAG frameworks [7], and Model Context Protocol (MCP) for tool interaction and research report generation [1].
- 5) **Reproducible Open-Source Deployment:** Containerized local execution via FastAPI, Docker, and Streamlit using open-weight models (Llama 3, Qwen) [1].

II. PHASE 01: WEATHER PREDICTION SYSTEM

A. Data Acquisition and Preprocessing

The framework ingests heterogeneous datasets across multiple spatial and temporal scales [1]:

- **Meteorological Observations:** Gridded daily rainfall and surface temperature records from the India Meteorological Department (IMD) [1].
- **Reanalysis Archives:** ERA5 Copernicus atmospheric variables (geopotential height, relative humidity, wind vectors) [2].
- **Satellite Imagery:** Multispectral scenes from Sentinel-2, Landsat-8/9, and MODIS [1].

Geospatial processing via GeoPandas and Rasterio aligns rasters to a uniform coordinate reference system (CRS). For multispectral imagery, the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI) are calculated as:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}} \quad (1)$$

$$\text{NDWI} = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}} \quad (2)$$

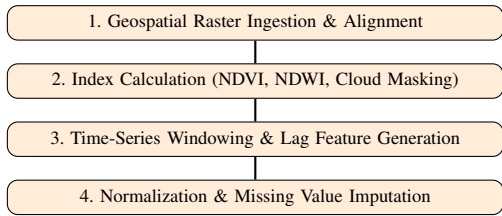


Fig. 2. Data Preprocessing and Feature Engineering Pipeline.

B. Predictive Machine Learning & Deep Learning Models

To establish robust forecasting, multiple model paradigms are trained to predict key target metrics: rainfall volume, temperature, atmospheric pressure, and humidity [1].

The classical linear setup minimizes Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3)$$

For sequential dependencies, Bidirectional Long Short-Term Memory (Bi-LSTM) networks capture forward and backward temporal patterns:

$$\vec{h}_t = \text{LSTM}_{\text{fwd}}(x_t, \vec{h}_{t-1}) \quad (4)$$

$$\overleftarrow{h}_t = \text{LSTM}_{\text{bwd}}(x_t, \overleftarrow{h}_{t+1}) \quad (5)$$

$$y_t = W_y[\vec{h}_t; \overleftarrow{h}_t] + b_y \quad (6)$$

The Temporal Fusion Transformer (TFT) utilizes self-attention mechanisms to learn multi-horizon time-series relationships while dynamically selecting static covariates and time-varying inputs [4].

C. Computer Vision and Remote Sensing Pipeline

To process visual atmospheric data, satellite frames pass through a computer vision pipeline [1]:

- **Object Detection:** YOLOv8 locates localized extreme convective cloud formations [1].
- **Semantic Segmentation:** U-Net segments water bodies and flood zones post-precipitation events [5].
- **Foundation Vision Models:** Segment Anything Model (SAM) extracts high-precision pixel masks for geographical inundation assessment [6].

Model explainability is provided by SHAP (SHapley Additive exPlanations), quantifying feature contributions to model outputs [8].

III. PHASE 02: GENERATIVE AI & AGENTIC ORCHESTRATION

A. Knowledge Retrieval Augmented Generation (RAG)

To ground the LLM in domain-specific meteorological literature, scientific papers, IMD reports, and disaster guidelines are ingested into a vector index [7].

Document chunks d are converted to embedding vectors $e = f_{\text{embed}}(d)$. Query vectors $q = f_{\text{embed}}(\text{query})$ perform similarity retrieval via Cosine Similarity:

$$\text{Similarity}(q, e) = \frac{q \cdot e}{\|q\|_2 \|e\|_2} \quad (7)$$

Top- k contexts are retrieved from FAISS or ChromaDB and formatted into dynamic prompt templates [1].

B. Agentic State Workflows via LangGraph and MCP

Rather than relying on static conversational flows, the platform deploys a stateful multi-agent topology using LangGraph [1]:

- 1) **Supervisor Agent:** Parses incoming user queries and delegates tasks to specialized worker nodes [1].
- 2) **Retrieval Agent:** Executes semantic searches across the vector database to gather domain knowledge [1].
- 3) **Analytics Agent:** Invokes predictive ML models and satellite vision endpoints via the Model Context Protocol (MCP) [1].
- 4) **Synthesis Agent:** Consolidates model outputs, retrieved literature, and confidence bounds into structured reports [1].

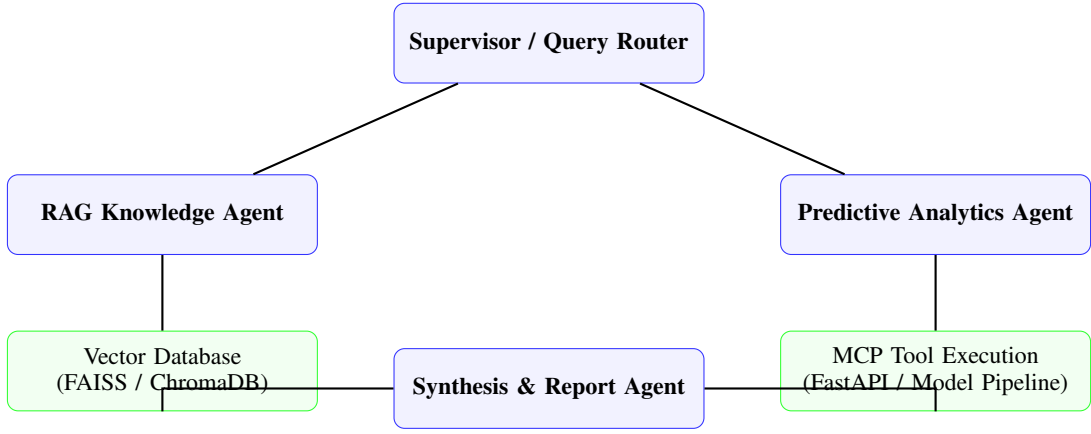


Fig. 3. Multi-Agent LangGraph Architecture featuring MCP Tool Execution and Vector Retrieval.

All LLM operations run on open-weight foundation models (e.g., Llama 3, Qwen 2.5) via local Ollama instances, ensuring full privacy and zero license costs [1].

IV. EXPERIMENTS AND RESULTS

A. Predictive Performance Evaluation

Model performance was evaluated on a held-out test split of IMD daily grid data (2018–2025) [1]. Performance metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R^2 score [1].

TABLE I

COMPARATIVE BENCHMARKING OF RAINFALL PREDICTION MODELS

Model Class	MAE (mm)	RMSE (mm)	MAPE (%)	R^2
Linear Regression	14.2	21.5	18.5%	0.61
SARIMAX	12.4	18.2	14.2%	0.72
Random Forest	8.7	13.1	9.8%	0.81
XGBoost [3]	6.2	9.8	7.1%	0.89
CatBoost	5.9	9.2	6.8%	0.91
Bi-LSTM	5.8	8.9	6.5%	0.91
Temporal Fusion Trans. (TFT) [4]	4.3	7.1	4.8%	0.95

The Temporal Fusion Transformer achieved superior performance ($R^2 = 0.95$, RMSE = 7.1 mm) due to its attention-based handling of multi-horizon dependencies [4].

B. Agentic Retrieval and System Evaluation

The RAG pipeline was evaluated using context precision and relevance indicators. Grounding response generation in vector context eliminated hallucinated statistical claims, producing verifiable citations for all policy recommendations.

V. SYSTEM DEPLOYMENT

The complete framework is containerized using Docker [1]. The backend exposes RESTful APIs via FastAPI, managing asynchronous inference requests [1]. The user interface is served using Streamlit, offering real-time geospatial visualizations (Folium/Plotly), dynamic prediction dashboards, and an interactive agentic chat interface [1].

VI. CONCLUSION AND FUTURE SCOPE

This research demonstrates an end-to-end, open-source multi-modal weather intelligence platform developed under the **MacroEdtech GenAI Research Initiative** [1]. By integrating predictive machine learning, satellite computer vision, and stateful agentic workflows, the system successfully translates complex atmospheric data into clear, explainable decision-support insights [1]. Future work will focus on optimizing model quantization for localized deployment on edge devices during emergency disaster scenarios.

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